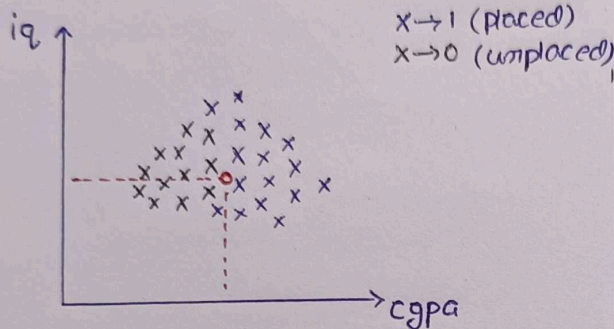


KNN

K-Nearest Neighbour:

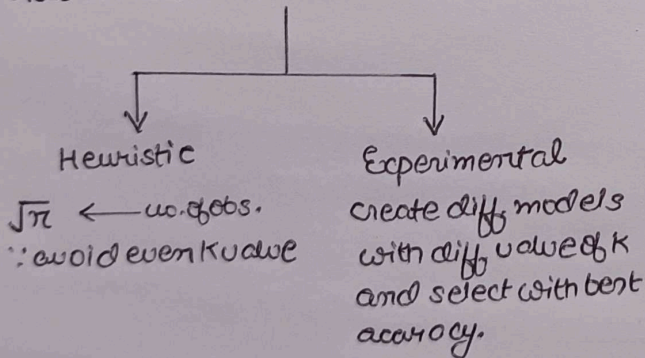
⇒ Basic intuition:

Suppose we have a dataset of students as CGPA, IQ, placed? and we get a new dataset to predict if it's placed or unplaced, using KNN.



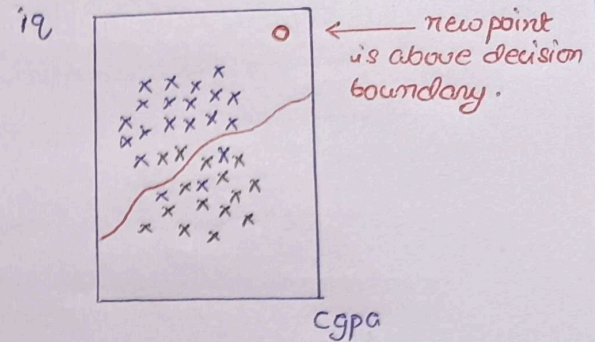
We will first select 'K', say $K=3$ (3 nearest neighbour). Then we find euclidean distance of new point from all given points and then arrange them in ascending order. Then, by using concept of majority count we select first three distances and whoever be in majority be the predicted result, say 1 or 0. Then '1' is majority and is the result.

⇒ How to select 'K'?



⇒ Decision surface: It's a tool used to create a rough idea of decision boundary of classification problems.

eg.,



⇒ Overfitting in KNN: (beg of high variance)
It's mainly because of small value of K and because of outliers.

⇒ Underfitting in KNN:
It's mainly beg of very large value of K.

⇒ Limitations of KNN:

- (i) For large datasets as it's a type of lazy learning technique.
- (ii) High dimension data as it leads to curse of dimensionality.
- (iii) It doesn't work well with outliers.
- (iv) Non homogeneous feature scale.
- (v) Imbalanced dataset.
- (vi) If we are using for inference and not for prediction.